



Hydraulic Analysis Report - T-101 : Internals-1@Main Tower@COL1

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Simulation case: C:\Users\C6H6_\Downloads\ANILINA_99.hsc

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User Name: ESIQIE

Job Code:

Project: T-101

Description: TORRE DE DESTILACION T-101

1. Column Internals Summary

Summary

Property	Value	Units
Number of Trayed/Packed stages	6	
Total height	3.658	m
Total head loss (Hot liquid height)	437.5	mm
Total pressure drop	40.89	mbar
Number of sections	1	
Number of diameters	1	
Total residence time	8.488	seconds

Sections

Section	Start	End	Diameter (m)	Section Height (m)	Internals Type	Tray or Packing Type	Section Pressure Drop (mbar)	% Approach to Flood	Limiting Stage
CS-1	1_Main Tower	6_Main Tower	0.5044	3.658	Trayed	Sieve	40.89	80.0003	6_Main Tower

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2. Feed/Draw Summary

Feed/Draw Summary

Stage Number	Stream	Feed/Draw
1_Main Tower	Reflux	Feed
	ORG-110	Feed
	To Condenser	Draw
6_Main Tower	Boilup	Feed
	To Reboiler	Draw

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3. CS-1

3.1. Tray Geometry

Section

Property	Value	Units
Tray type	SIEVE	
Diameter	0.5044	m
Tray spacing	0.6096	m
Number of passes	1	
Hole diameter	0.0127	m
Hole area / Active area	0.1	
Number of Holes	126	
Deck gauge thickness	10 GAUGE	
Deck gauge thickness value	3.404	mm
Cross-sectional area	0.1998	m2
Active area	0.1598	m2
Net area	0.1798	m2

Downcomer geometry

Property	Side	Units
Downcomer clearance	0.0381	m
Downcomer width top	0.07892	m
Downcomer width bottom	0.07892	m
Downcomer area top	0.01998	m2
Downcomer area bottom	0.01998	m2

Weir geometry

Property	Side	Units
Weir height	0.0508	m

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Property	Side	Units
Weir length	0.3665	m

Panels

Property	A	Units
Flow path length	0.3465	m
Bubbling area	0.1598	m2

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3.2. Design Parameters

Design Parameters

Property	Value	Units
% Jet flood and downcomer choke flood for design	80	
Minimum downcomer area / Total tray area	0.1	
Downcomer design basis		
Maximum acceptable pressure drop	2.5	kPa
Maximum % jet flood	100	
Maximum % downcomer backup	100	
Maximum % liquid entrainment	10	
Minimum weir loading	4.471	m3/h-m
Maximum weir loading	117.4	m3/h-m
Warning status (% to limit)	10	
System foaming factor	1	
Aeration factor multiplier	1	
Over design factor	1	
Overall section efficiency	1	
Weep method	Hsieh	
Jet flood method	Glitsch6	
Maximum downcomer loading method	Glitsch	
Dry pressure drop multiplier	1	